AN END-TO-END BIG DATA PIPELINE

Detecting IoT Malware at Network Scale.

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BRIEF

Can we classify malicious IoT network traffic from raw Zeek connection logs?

25.0M
NETWORK CONNECTIONS

12
IOT CAPTURE SOURCES

7
MALWARE BEHAVIOUR CLASSES

0.997
RANDOM FOREST · F1

A Hadoop / Spark pipeline that ingests 25M Zeek records from PostgreSQL, partitions them in Hive, and trains distributed multiclass classifiers for seven malware behaviours.

STAGES OF WORK

From CSV to dashboard, in four reproducible stages.

STAGE 01	STAGE 02	STAGE 03	STAGE 04
Ingestion	**Storage & EDA**	**Predictive Analytics**	**Dashboard**
PostgreSQL + Sqoop import to HDFS as Parquet/Snappy.	Hive partitioned + bucketed tables. Seven Tez/HiveQL insights.	Spark ML pipeline — RF and Multinomial LR with grid-search CV.	Apache Superset — data characteristics, EDA insights, model results.

CLUSTER

Innopolis Hadoop · 3 nodes · YARN · Hive + Tez · Spark on YARN.

STACK

Python 3 · PySpark · Sqoop · Beeline · psycopg2 · Apache Superset.

REPRODUCIBILITY

Single entry point `main.sh` — every stage is idempotent and re-runnable.

01

Data Collection & Ingestion.

12 CAPTURE FILES. TWO RELATIONAL TABLES. ONE SQOOP IMPORT.
PARQUET ON HDFS, SNAPPY-COMPRESSED, READY FOR HIVE.

SOURCE

Real IoT-device captures from the Aposemat IoT-23 corpus.

12 selected captures of *Mirai, Hide-and-Seek, Hakai, Torii, Trojan, Okiru, Kenjiro* and benign baselines — converted to Zeek `conn.log` format with binary & multi-class malware labels.

VOLUME & SHAPE

ROWS	25,011,247 network connections
CAPTURES	12 distinct IoT-device traces
FEATURES	23 per-connection Zeek fields
TARGET	7-way `label` (Benign + 6 attack types)

1.1 RELATIONAL SCHEMA

Two tables, one foreign key.

CAPTURES

- **capture_id** int (PK)
- capture_name varchar(128)
- source_file varchar(255)
- created_at timestamptz

NETWORK_CONNECTIONS

- **connection_id** bigint
- capture_id int (FK)
- ts timestamptz
- id_orig_h / id_resp_h inet
- proto · service · conn_state varchar
- orig_bytes / resp_bytes bigint
- orig_pkts / resp_pkts bigint
- history · duration · ports …
- label · detailed_label varchar

Stage table `stg_network_connections` ingests raw text via COPY; type casts (`inet`, `to_timestamp`, regex-cleaned ints) happen in the SQL transform.

1.2 INGEST PIPELINE

Four steps. One reproducible load.

STEP 01	STEP 02	STEP 03	STEP 04
Collect	**Load**	**Validate**	**Sqoop**
gdown pulls 12 capture CSVs from Google Drive into `data/`.	COPY into staging, then transformed insert into typed PostgreSQL tables.	7 SQL assertions on row counts, label coverage and FK integrity.	Both tables exported to HDFS as Parquet + Snappy via single-mapper queries.

1.3 STORAGE BENCHMARK

We compared four format / codec pairs.

FORMAT	CODEC	WRITE (S)	SCAN (S)
parquet	snappy	17.7	6.76
parquet	gzip	18.0	6.64
avro	snappy	17.0	9.32
avro	bzip2	19.0	8.36

Parquet + Snappy chosen — best scan/throughput trade.

1.4 SQOOP COMMAND (EXCERPT)

```
# network_connections → HDFS
sqoop import \
  --connect jdbc:postgresql://hadoop-04/team30_projectdb \
  --compression-codec snappy --compress \
  --as-parquetfile \
  --target-dir /user/team30/project/
    warehouse/network_connections \
  --m 1 \
  --query "SELECT … FROM
    network_connections WHERE \$CONDITIONS"
```

Single mapper avoids skew on the auto-incremented PK; UTC-formatted timestamps survive the round-trip.

02

Storage & EDA.

EXTERNAL PARQUET → BUCKETED CAPTURES + LABEL-PARTITIONED CONNECTIONS. SEVEN HIVEQL QUERIES RUN ON TEZ; RESULTS MATERIALISED AS PARQUET RESULT TABLES.

2.1 OPTIMISED TABLES

Two physical layouts — one for joins, one for label scans.

EXT	`captures_ext` & `network_connections_ext` over Sqoop output.
BUC	`captures_bucketed` — CLUSTERED BY (capture_id) INTO 4 BUCKETS.
PART	`network_connections_part` — PARTITIONED BY (label).
FMT	All managed tables stored as PARQUET on HDFS.

WHY Label is the dominant filter in EDA & ML — partitioning by it lets every analytic query prune to a fraction of the data.

2.2 ENGINE & SETTINGS

HiveQL on Tez with dynamic partitioning.

```
SET hive.execution.engine=tez;
SET hive.exec.dynamic.partition=true;
SET hive.exec.dynamic.partition.mode=nonstrict;
SET hive.enforce.bucketing=true;

-- bucketed insert
INSERT OVERWRITE TABLE captures_bucketed
SELECT capture_id, capture_name, source_file,
  CAST(created_at AS TIMESTAMP)
FROM captures_ext;

-- partitioned insert (label dynamic)
INSERT OVERWRITE TABLE network_connections_part
PARTITION (label) SELECT … , label
FROM network_connections_ext;
```

7 EDA queries materialise `qN_results` Parquet tables and stream CSV exports through Beeline for the dashboard.

01

A SKEWED BUT RICH TARGET

Benign and Malicious dominate — six rare attack types remain useful.

The corpus is neither balanced nor degenerate: 35% benign, 28% generic Malicious, 23% DDoS, 14% port-scan, plus three rare classes that the multiclass models still need to handle.

Benign label = "Benign"		35.11%
Malicious label = "Malicious"		28.21%
DDoS label = "Malicious DDoS"		23.10%
Port scan PartOfAHorizontalPortScan		13.54%
C&C / Attack / FileDownload 3 rare classes combined		0.05%

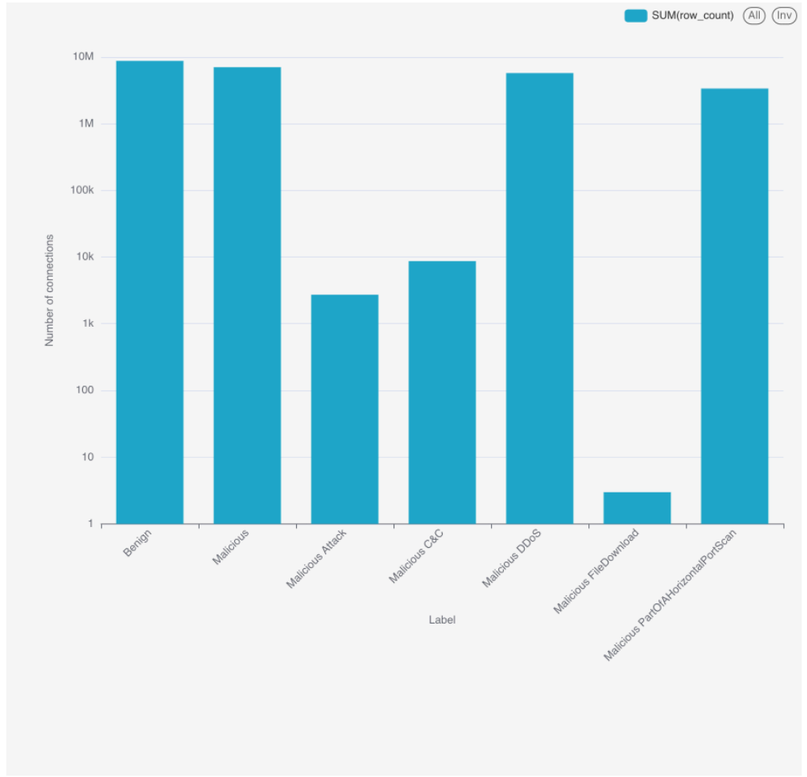


FIG. 01 · CLASS SHARE OF NETWORK_CONNECTIONS_PART. THE LONG TAIL OF RARE ATTACK TYPES IS THE KEY REASON THE PROJECT IS TREATED AS MULTICLASS, NOT BINARY.

02

ALMOST EVERYTHING IS TCP

Protocol mix per label.

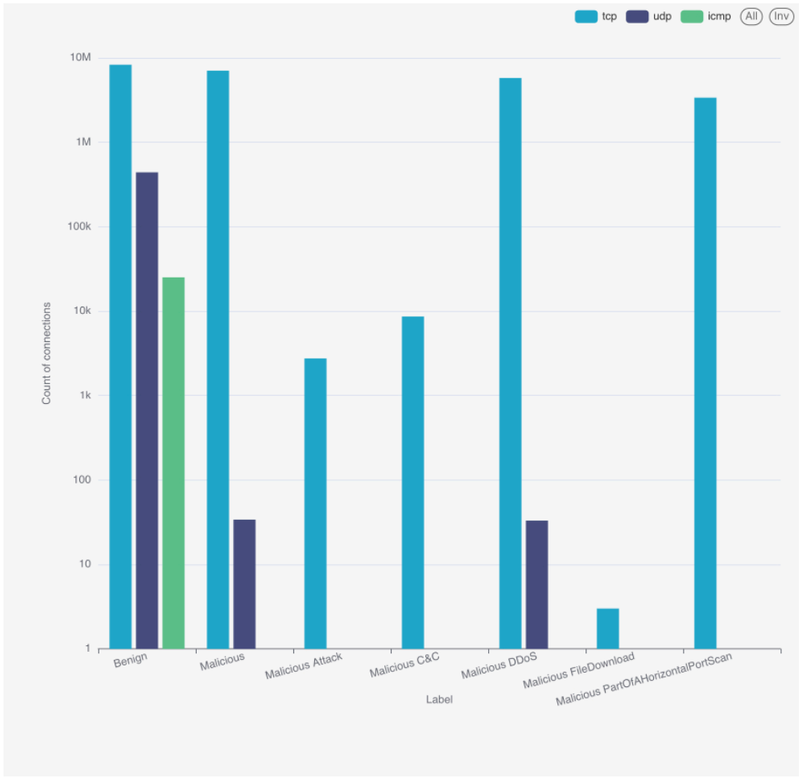


FIG. 02 · UDP APPEARS ALMOST ONLY FOR BENIGN DNS AND ICMP-ONLY FLOWS; NEARLY EVERY MALICIOUS CONNECTION IS TCP.

05

FAILED HANDSHAKES EVERYWHERE

Connection state per label.

LABEL	DOMINANT CONN_STATE	SHARE
Benign	S0 (no reply)	99.3%
Malicious	S0	99.7%
Malicious DDoS	OTH / RSTOS0	99.99%
Malicious C&C	S0 + S3	88%
Port scan	S0	100%

A SYN-only flood (S0) is the universal signature — benign traffic on this network is dominated by IoT devices repeatedly failing to reach a server.

03 VOLUMETRIC ATTACKS VS. SILENT SCANS

Average bytes per connection by label.

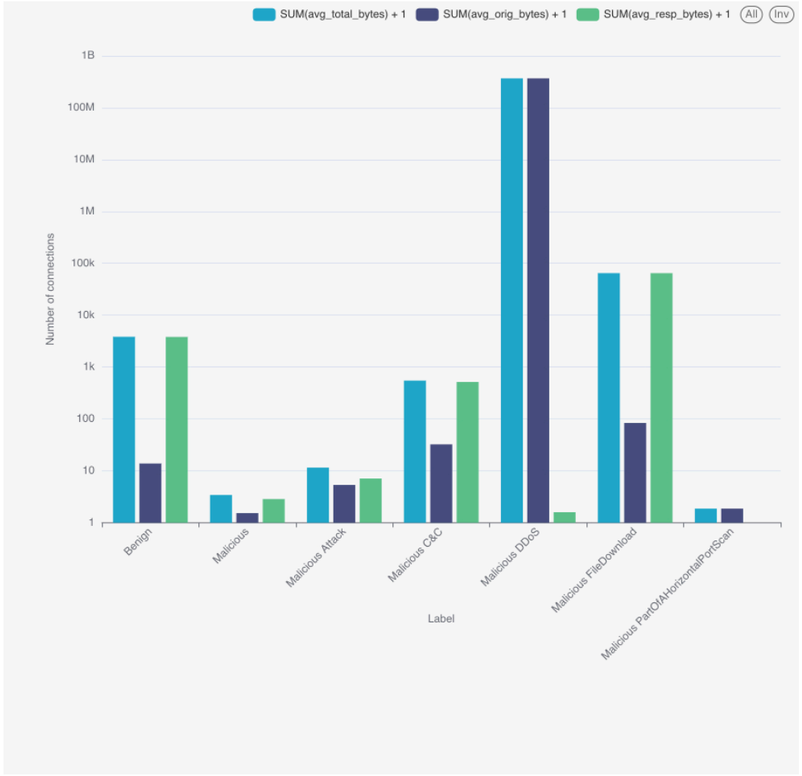


FIG. 03 · DDOS AVERAGES ≈ 3.7×10^8 ORIG BYTES / FLOW — SIX ORDERS OF MAGNITUDE OVER BENIGN. PORT-SCAN AND GENERIC MALICIOUS CARRY < 1B PER CONNECTION ON AVERAGE.

07 PACKET-RATE SIGNATURES

Average packets per connection.

LABEL	AVG ORIG PKTS	AVG RESP PKTS
Benign	3.90	0.006
Malicious	1.10	0.033
Malicious DDoS	48.10	0.0004
C&C	495.4	8.15
Attack	5.34	3.40
Port scan	4.00	0.00

Long-lived C&C sessions stand out: ~500 pkts out, with a real (8 pkt) reply — very different from one-shot scan or flood traffic.

04

CAPTURES CLUSTER BIMODALLY

Almost all malicious or almost all benign — nothing in between.

8 of the 12 captures are > 79% malicious; 4 captures are < 11% malicious. There is no middle ground — each PCAP was deliberately recorded around a single behaviour.

CAPTURE	TOTAL FLOWS	MALICIOUS RATIO
Capture-60	3,581,028	99.9%
Capture-9	6,378,293	99.6%
Capture-48	3,394,338	99.9%
Capture-3	156,103	97.1%
Capture-1	1,008,748	53.5%
Capture-44	237	11.0%
Capture-20 / 21 / 42	~3,500 each	<1%

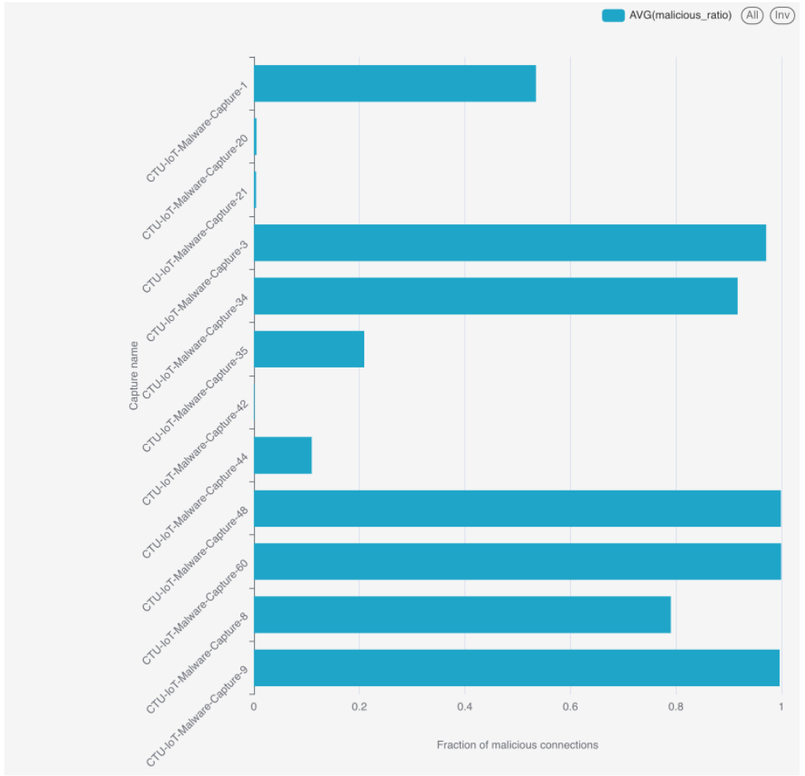


FIG. 04 · TREAT CAPTURES AS *STRATA*, NOT INTERCHANGEABLE ROWS — RANDOMLY SPLITTING AT THE ROW LEVEL CAN LEAK WHOLE DEVICES INTO BOTH TRAIN AND TEST.

06 TELNET IS STILL THE WOUND

Port 23 is the single most attacked destination.

Generic Malicious traffic (4M flows) and the entire port-scan class (3.4M flows) target Telnet (`tcp/23`) — the classic Mirai-family entry point. DDoS is bimodal between 62336 (Mirai-style C&C echo) and HTTP 80.

LABEL	PORT	FLOWS
Malicious	23	4,055,327
DDoS	62336	3,578,391
Port scan	23	3,386,119
Malicious	81	2,571,979
DDoS	80	1,125,806
DDoS	992	1,008,351
C&C	6667	6,706

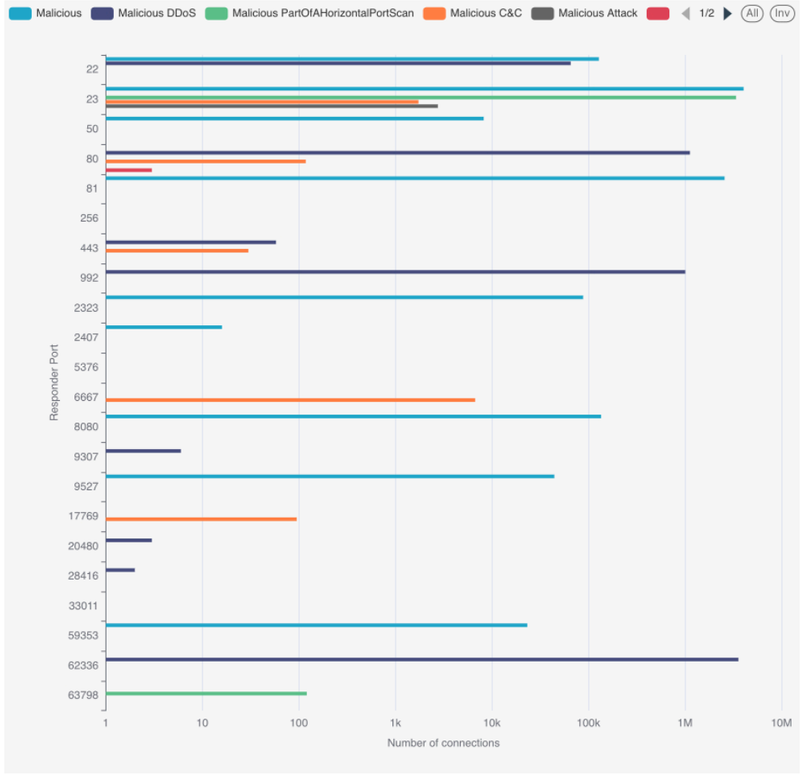


FIG. 06 · TARGET-PORT HISTOGRAMS PER NON-BENIGN LABEL. PORT 23 DOMINANCE IS A STRONG, LEARNABLE SIGNAL — IT IS ONE OF THE HIGHEST-IMPORTANCE FEATURES IN THE TRAINED RANDOM FOREST.

03

Predictive Analytics.

A 7-CLASS SUPERVISED PROBLEM ON 25M ROWS. CYCLICAL TIME ENCODING, CHISQ FEATURE SELECTION, TWO CROSS-VALIDATED MODELS — ALL DISTRIBUTED ON YARN.

3.1 PIPELINE

From raw Hive rows to a single scaled feature vector.

STEP 01	STEP 02	STEP 03	STEP 04	STEP 05
Time encode	**Impute**	**One-hot**	**Select**	**Assemble + scale**
Custom `CyclicalTimeTransformer` — sin/cos for month, day, hour, minute, second.	Fill nulls per dtype: "unknown" for strings, 0/false for numerics, drop `tunnel_parents`.	`StringIndexer + OneHotEncoder` for proto, service, conn_state, history.	`ChiSqSelector(fpr=0.1)` over categorical block — keeps only label-correlated dummies.	`VectorAssembler` + `StandardScaler` → final `features` column.

NUMERIC FEATURES

duration · orig_bytes · resp_bytes · missed_bytes · pkts (orig/resp) · ip_bytes · ports · year · local_orig · local_resp.

CATEGORICAL (AFTER CHISQ)

proto · service · conn_state · history — selected on training fold only to avoid leakage.

SPLIT

60% / 40% random split, seed = 42. Final train/test persisted as Parquet on HDFS for both training and evaluation.

3.2 TWO MODEL FAMILIES

A non-linear baseline against a linear baseline.

M1 **Random Forest** — numTrees ∈ {30, 60} × maxDepth ∈ {4, 6}. Bagging handles the long tail of rare classes naturally.

M2 **Multinomial Logistic Regression** — regParam ∈ {0.01, 0.1} × elasticNet ∈ {0.0, 0.5}, maxIter = 50.

CV 2-fold cross-validation, parallelism = 3, optimised on multiclass `f1`.

CLUSTER

Each `spark-submit`: 6 executors × 3 cores × 6 GB on YARN. Train data `.persist(MEMORY_AND_DISK)` to amortise CV folds.

3.3 BEST RANDOM FOREST

Hyperparameters chosen by CV.

HYPERPARAMETER	BEST VALUE
numTrees	60
maxDepth	6
impurity	`gini`
featureSubsetStrategy	`auto`

3.4 BEST LOGISTIC REGRESSION

HYPERPARAMETER	BEST VALUE
regParam	0.01
elasticNetParam	0.0
family	`multinomial`
maxIter	50

3.5 COMPARISON

Both models clear F1 = 0.99 — Random Forest slightly ahead.

After `ChiSqSelector` trims uninformative dummies and `StandardScaler` harmonises feature scales, the multinomial linear baseline is genuinely competitive on this problem.

MODEL	ACCURACY	PRECISION	RECALL	F1
Random Forest	0.9974	0.9974	0.9974	0.9973
Logistic Regression	0.9943	0.9939	0.9943	0.9941

Metrics computed via PySpark `MulticlassClassificationEvaluator`; F1 weighted by class support.

3.5 VISUAL RANKING

Bars scaled by F1 score.

RF reaches **F1 = 0.997** on a 7-class, 25M-row problem; LR sits 0.003 behind at **0.994** — a strong baseline rather than a straw man.

Predictions persisted to HDFS as CSV (`project/output/modelN_predictions/`) and replayed by `ml_evaluation.py` — evaluator is decoupled from training.

04

Dashboard.

A SINGLE SUPERSET DASHBOARD BACKED BY THE SEVEN Q*_RESULTS HIVE TABLES AND THE MODEL EVALUATION CSV. PUBLIC, REPRODUCIBLE, REFRESH-ON-DEMAND.

4.1 DASHBOARD SECTIONS

From raw scale to model verdict, in five panels.

A **Data characteristics** — 25M rows, 23 features, 7 classes, 12 captures.

B **Class balance** (q1) — donut and percentage list.

C **Behaviour signatures** (q2, q3, q5, q7) — protocol, bytes, conn_state, packets.

D **Top targets** (q6) — port histograms per attack type.

E **Capture strata** (q4) — bimodal maliciousness ratios.

F **Model performance** — RF vs. LR comparison from `evaluation.csv`.

4.2 WIRING

Backed entirely by managed Hive tables.

```
-- Superset → Hive 2 (HiveServer2)
jdbc:hive2://hadoop-03.uni.innopolis.ru:10001
    USE team30_projectdb;

-- One dataset per chart
SELECT label, row_count, share_fraction
  FROM q1_results;

SELECT model, accuracy, precision, recall, f1_score
  FROM evaluation_csv;
```

REPRODUCIBILITY CONTRACT

Re-run `main.sh` on the cluster — every Hive result table and the evaluation CSV are rebuilt; the dashboard refreshes against them with no manual edits.

CI · THREE LAYERS

Heavy work stays manual; the per-commit loop stays fast.

The full pipeline downloads ~3.39 GB and competes for YARN — never on a push. CI runs only what's safe and quick.

LAYER	TRIGGER	WHAT
1 · Lint / unit	push / PR	ShellCheck, ruff + pylint, `pdflatex`, bats unit
2 · Cluster smoke	manual	Postgres, HDFS, Hive, Spark eval re-run
3 · Full pipeline	manual	`bash main.sh` end-to-end

LAYER 2 — CLUSTER RUNNER

```
bash scripts/run_tests.sh smoke 02_postgres
```

CI · LIVE STATUS

Four jobs in parallel, ~1m wall-time.

Green: `ShellCheck` · `ruff + pylint` · `bats unit` · `Build report.pdf`.

Earlier run on the same PR — `ruff + pylint` caught a regression in seconds, the other three jobs surfaced partial feedback in parallel.

WHAT WE BUILT

A reproducible Hadoop pipeline for IoT malware classification — from CSV to model to dashboard.

SCALE	25M Zeek rows ingested to Hive in < 20 s with Parquet/Snappy.
SIGNAL	7 EDA insights expose class skew, port-23 dominance, S0-flood signature.
MODEL	Both RF and LR clear F1 = 0.99 across 7 classes; RF leads by ~0.003.
SHIP	Single `main.sh` entry point + public Superset dashboard.

REFLECTIONS & NEXT STEPS

What we'd change in v2.

LESSONS

Capture-level bimodality (q4) is a real risk for row-level random splits — group-aware splitting per `capture_id` is the right next iteration. RF leads LR by only ~0.003 F1, so the next ensemble worth trying is Gradient-Boosted Trees rather than more tuning of either head.

NEXT STEPS

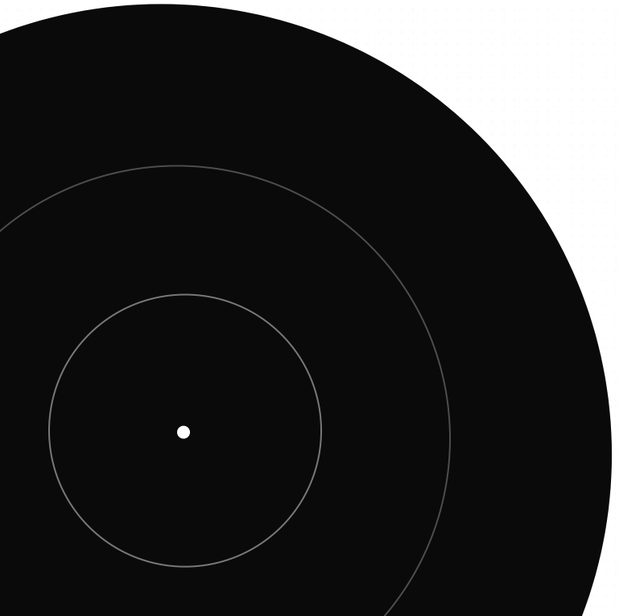
Gradient-boosted trees · per-capture cross-validation · streaming variant via Spark Structured Streaming on the same Hive partitions · per-class precision/recall in the dashboard.

FINAL POSITION

Big-data tools were the right hammer here — Hive partitioning + Spark CV made the 25M-row, 7-class job tractable on a 3-node cluster.

THANK YOU

Questions welcome.



A Hadoop / Spark pipeline that turns 25 million raw IoT connections into a single, reproducible verdict — and a dashboard maintainers can actually read.